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Efficiency in thermoelectric generators based on Peltier cells

Luigi O. Freire^{*}, Luis M. Navarrete, Byron P. Corrales, Jessica N. Castillo

Universidad Técnica de Cotopaxi, Latacunga, Ecuador

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Abstract

Obtain energy from thermoelectric generators based on Peltier cells, using natural and/or artificial heat sources in order to have a new ecological tool that allows the generation of clean energy. The present study is focused on a bibliographic research to determine the parameters that intervene in the validity of energy conversion and it has as an objective to analyze the efficiency of Peltier cells, in a didactic process of heat transference, which provides the necessary conditions to a thermoelectric generator based on Peltier cells. Experiments are carried out based on the changes in the process conditions and the transfer function of process is obtained to achieve efficiency in energy production.

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1. Introduction

Thermoelectric generators directly convert heat into electricity, this energy source is an effective alternative that focuses on reducing environmental impact [1]. Analyzing the possibility of using waste heat from processes as a promising source of electrical energy [2]. Where the heat induces the circulation of an electric current by flowing from the gas burner through the thermoelectric module [3]. A Thermoelectric Module (TEM) with Peltier cells is a specific electronic component that works as a small heat pump, is formed by alternative semiconductor thermocouples of type n and p [4,5]. When supplying the thermoelectric modules with electricity, they act as refrigerators, while when supplied at a temperature difference they generate a certain voltage [5]. Considering that thermoelectric devices can be associated with the operation of four physical effects: The Seebeck effect, the Peltier effect, the Thomson effect and the Joule effect [6,7].

The use of thermoelectric generators (TEG), has its origin from the year 1909 with the German scientist Edmund Altenkirch who was interested in the calculation of its efficiency, use and precision [8]. The research of Peltier Cells as TEM is based on the Seebeck effect. Thermoelectric materials convert heat energy into electrical energy directly

^{*} Corresponding author.

E-mail address: luigi.freire@utc.edu.ec (L.O. Freire).

Nomenclature

TEM	Thermoelectric module
TEG	Thermoelectric generators
TE	Thermoelectric
ΔT	Temperature gradient
T_h	Hot temperature
T_c	Cold temperature
RTE	Electrical resistance
R_e	External load resistance
P	Power transfer
η	Efficiency

by this effect [2]. It consists in the creation of a potential difference in the joints of two different metallic materials, as a result of application of a temperature gradient in magnetic materials, including insulators [9].

The modern development of Peltier cells began in 1990 when it built a U-shaped cell, five pairs of these cells provide 30 mW with the flame of a candle, enough to light a radius, in 2004 Toshiba Co. developed a cell that produces 3 mW/cm² for application in automobiles, in 2006 Yamaha built a cell that provides 0.5 mW/cm² with a temperature difference of 100 °C, in 2009 KELK Ltda. obtained a cell of 1 mW/cm². As of 2011, research has focused on building multilayer modules, flexible cells, tubular modules and ink-jet printed modules for special applications that, when properly combined, can reach powers of hundreds of watts [10].

The efficiency in TEG is defined as the ratio between the heat entering the system and the work resulting from it. To obtain efficiency in the generation of energy, the temperature gradient must be high and properly transferred to the surface of cells, so the materials must have a high degree of thermal conductivity [11]. Another factor is the oscillation of temperature and the phase of these oscillations between the hot and cold surfaces [12]. Factors that may negatively affect efficiency are the low figure of merit (Z) of materials used in the manufacture of cells [13] and the inadequate adjustment of thermoelectric systems (TE) [11].

In order to analyze the efficiency, a didactic module of heat transfer is used, from this the mathematical model has been obtained, this simplifies and allows to have an approximate idea of response of process in different conditions of temperature gradients. A comparison of results obtained when using a commercial Peltier cell versus the manufacturer's characteristic curves is presented. The research culminates with the determination of ideal conditions under which the process operates in areas of maximum efficiency.

The results can be applied to the generation of electrical energy from natural thermal sources such as: solar radiation, geothermal sources, etc. As well as from artificial thermal sources, combustion engines, incinerators and vehicles, processes in which two thirds of energy is lost in the form of heat [2].

The research was focused: Section 2 presents the structure of heat transfer module and the Peltier cells used in this research. Section 2.2 characterizes the Peltier cells as TEG and describes the process for obtaining the mathematical model of heat transfer module and the TEG. Finally, in Section 3, the experimental results of implementation of research are analyzed. Section 4 refers to the conclusions of data of thermoelectric generators based on Peltier cells.

2. System structure

2.1. Heat transfer system

Heat sources are little used for the generation of other types of energy in places that have natural thermal sources and in processes based on hydrocarbon combustion, the heat they produce is wasted. By means of a heat transfer process, it is desired to determine the efficiency of Peltier cells, under different conditions and compare them with the response curves.

The composition of Peltier generation system consists of a heat source to obtain a temperature gradient between the two sides of cells. A closed system is used to maintain the temperature on the hot side, while on the cold side there is a heat dissipater to dissipate the heat to the environment.

Fig. 1 presents the basic diagram of a heat transfer system, a 4500 W heater heats the water flow that circulates through the coil, two manual valves at the entrance and exit of process allow the flow to be controlled manually.

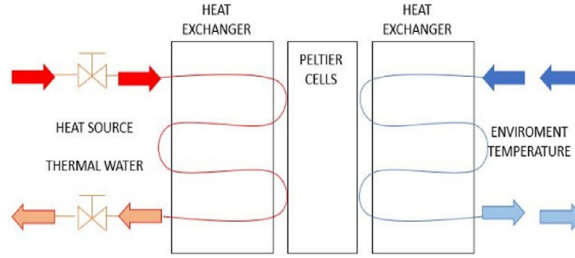


Fig. 1. Process diagram.

A metal plate has been placed in the heat exchanger to transfer the heat to the hot surface of cells; these have been physically arranged in the form of a matrix.

To find the appropriate configuration in the experimental case, the electrical connections of cells are arranged in (a) series, (b) parallel-series and (c) serial-parallel, as shown in Fig. 2.

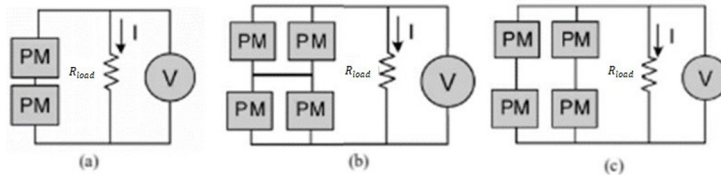


Fig. 2. Peltier cell configuration.

It can be shown that it is simple to understand the configuration of cells, these establish the voltage and current that will circulate through the load applying the sum of currents and voltages of circuits in series and parallel

2.2. Characterization of thermoelectric generator

The Seebeck effect governs thermoelectric generation based on Peltier cells. The voltage obtained at the terminals of a cell is given by (1) [7].

$$\Delta E = \alpha \Delta T \quad (1)$$

where α is called Seebeck coefficient, the same represents the voltage generated by each 1 K gradient this depends on the materials with which the cell is built. ΔT is the temperature gradient between each cell surface. However, α is not a constant and depends on the temperature gradient, therefore (2).

$$E = \int_{T_c}^{T_h} \alpha(T) dT \quad (2)$$

where T_h y T_c are the values of temperature gradient and T is the average $T_h + T_c$. Therefore, if the gradient tends to zero, α could become zero [2].

The current generated at the cell terminals, when a temperature gradient is applied, is given by (3).

$$I = \frac{\alpha \Delta T}{R_{TE} + R_e} \quad (3)$$

where, R_{TE} is the electrical resistance of TEG and R_e is the external load resistance. The power delivered to the load is given by (4).

$$P = I^2 R_e = \frac{S^2 \Delta T^2 R_e}{(R_{TE} + R_e)^2} \alpha T^2 \quad (4)$$

where for maximum power transfer, $R_{TE} = R_e$, therefore, the maximum power delivered to the load is given by.

$$P = \frac{\alpha^2 \Delta T^2}{4R_{TE \max}} \quad (5)$$

where the performance of materials is evaluated based on the figure of thermoelectric merit Z given in (1/K). This establishes the efficiency of a material in terms of heat to electricity conversion, when the temperature gradient is 1-degree Kelvin (K) and is expressed in

$$ZT = \frac{\alpha^2 \sigma}{\kappa} T \quad (6)$$

where, σ is the electrical conductivity and κ is the thermal conductivity. High values of α , σ and low values of κ , will allow to obtain a high ZT and therefore a high degree of efficiency.

The conditions of cell's environment are given by the temperatures on the hot surface, the cold surface and the side surface; the latter assumes that there is no heat transfer, so it is not considered for the calculation. The temperature on the hot side is represented as a sinusoidal function and is given by.

$$T_h = T_{h,\beta=0} + \beta_h \sin\left(\frac{2\pi T}{\lambda_h}\right) \quad (7)$$

where, the first term represents the temperature on the hot surface when the amplitude is equal to zero, β is the temperature amplitude (K), t time (min) and λ the period (min) [12]. The sinusoidal temperature of cold surface is given by.

$$T_c = T_{c,\beta=0} + \beta_c \sin\left(\frac{2\pi T}{\lambda_c} + \omega\right) \quad (8)$$

where the interpretation is homologous to the hot surface equation, ω is the phase angle of temperature between the two surfaces. It is experimentally demonstrated in [12] that the greater the phase angle ω , the greater the amplitude difference between T_h and T_c , therefore there will be greater efficiency in power generation. The efficiency of a TEG is expressed in.

$$\eta = \frac{(T_h - T_c)(\gamma - 1)}{T_c + \gamma T_h} \quad (9)$$

where, $\gamma = (1 + ZT)^{1/2}$ varies with the average temperature T [14]. The module efficiency is an important factor especially when the available heat is limited, the maximum efficiency conversion is given by [15,16].

$$\eta = \frac{P_{\max}}{\dot{Q}_{\text{hot}}} \quad (10)$$

where, $\dot{Q}_{\text{hot}}$ is the heat that flows on the hot surface of cell and $P_{\max}$ is measured under load conditions.

2.3. Workplace temperatures

The location chosen to take advantage of geothermal energy is Papallacta, this is a town and parish located in the Quijos canton of Napo province, in Ecuador; located 67 km east of city of Quito.

This population there are hot springs and an average temperature of 10 °C as shown in Fig. 3 where it indicates the minimum, average and maximum temperatures of Papallacta.

The thermal waters of Papallacta have a temperature that varies from 30 °C to 70 °C, these will help us to obtain the temperature difference and thus test the power generation of our Peltier cell bank.

3. Results and discussion

To experiment the efficiency of process, the system consists of a heat generator where the temperature gradient was raised to 75 °C, which determines the level of voltage that can be obtained at the output. The TEC1-12706 cells in the TEG teaching module consist of 104 couples in a mixed connection to obtain a voltage and current for applications in electrical equipment.

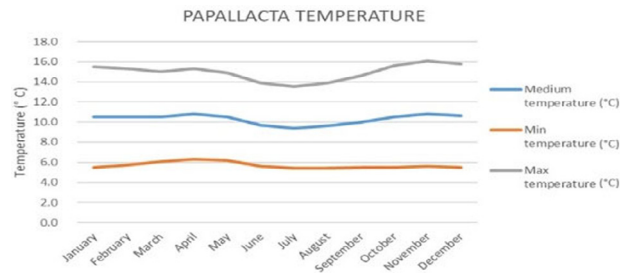


Fig. 3. Papallacta temperature.

The experimental behavior of thermodynamic system observed in Peltier cells shows that the electric current is a second degree function of temperature difference.

$$I = 0.0032 (\Delta T)^2 - 0.0613 \Delta T + 0.0713 \quad (11)$$

In order to determine the maximum efficiency of system, experimental conditions are established in the laboratory, with the ambient temperature being 24 °C and the maximum temperature reached 100 °C. It should be taken into account that the final temperature of cold surface reached 32 °C due to the effect of thermal conduction and remained at that value until the end of experiment.

Fig. 4 shows a completely linear relationship for the simple and approximately linear configuration from 50 mA in mixed configurations. The resulting power is a function of voltage and current generated according to each configuration.

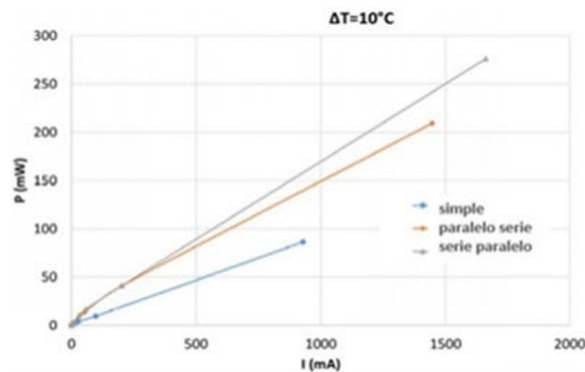


Fig. 4. Measured power of parallel-series combination versus current.

Fig. 5 the no-load voltage values were obtained for each ΔT , this data collection was carried out in a faster and more direct way, without any type of external resistance.

According to the data obtained, the greatest amount of voltage will occur when we have an ambient temperature of 10 °C and the temperature of thermal water of 70 °C, indicating that according to the trend of curve while the ambient temperature is lower the voltage that the Peltier cell will produce will be higher as shown in Fig. 6.

The efficiency of device has been found to increase with the difference in temperature to which it is exposed as shown in Fig. 6. From which we deduce that with a temperature difference of approximately 55 degrees the efficiency is around 2%.

These percentages seem insignificant at first glance, but the objective of plates Peltier is not that of generating energy, but another concept very similar to practice but fundamentally different, take advantage of it.

With this graph, it is not possible to extract a clear equation that defines the efficiency of device studied for each temperature difference, but it does give a very good idea clear of potential that Peltier plates have. Well it can be deduced that for the whole usable range of temperatures, the efficiency will be around these values.

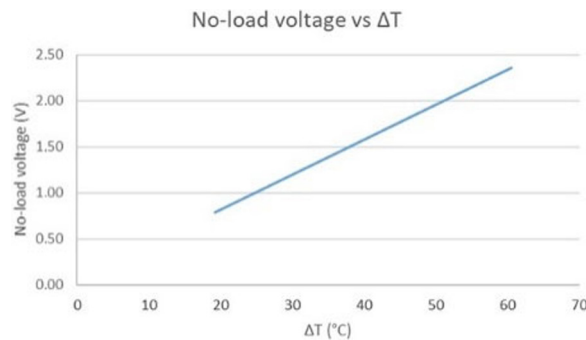


Fig. 5. No-load voltage vs. ΔT .

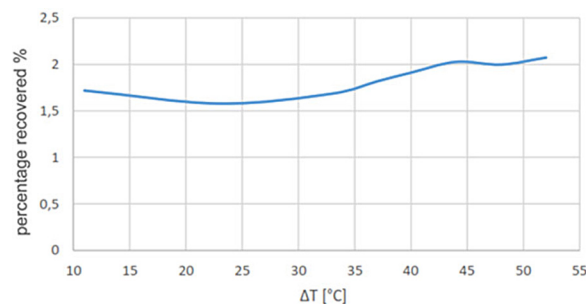


Fig. 6. Efficiency.

4. Conclusions

- It was possible to demonstrate in the investigation that the simulation showed the temperature gradients present in areas with thermal waters of mountain range, the same ones that can be used for the generation of electrical energy and its conversion in other forms in applications in electrical equipment of low power and/or storage.
- The experimental module of heat transfers and the TEG system allowed determining the ranges of generation temperature gradients. It allows relating and modifying the optimum working conditions in real applications, whether natural or artificial thermal sources.
- By determining the mathematical model of process, it is possible to establish the ideal conditions in which Peltier cells can be more efficient in real systems. If the temperature gradient is not sufficient, it can be used in combination with a collector system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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